SECURE AND UNBIASED QUANTUM ENERGY TELEPORTATION VIA QUANTUM OBLIVIOUS TRANSFER

A hybrid approach of Quantum Energy Teleportation protocol with Quantum Oblivious Transfer for secure and unbiased energy teleportation

Presented By

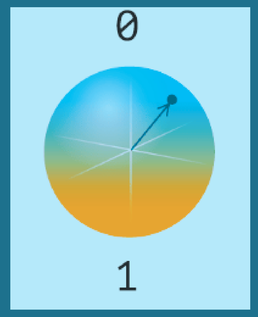
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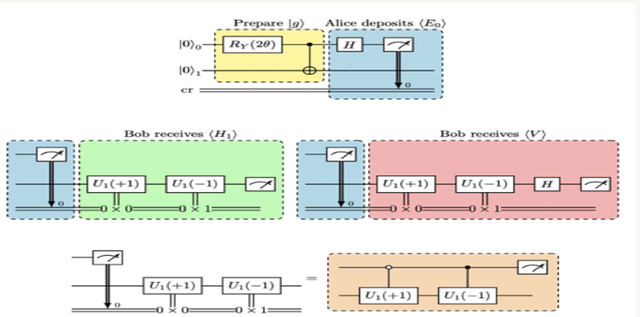


ABSTRACT

Quantum Energy Teleportation (QET) via Quantum Oblivious Transfer (QOT) represents a groundbreaking fusion of quantum information and energy transfer protocols. QET enables the transmission of quantum state energy between remote locations using local operations and classical communication, while QOT ensures secure information exchange where a sender transmits data obliviously to a receiver. By integrating QOT into QET, we achieve secure and efficient energy redistribution in quantum networks. This protocol has potential applications in quantum communication, cryptography, and energy-efficient quantum systems. Our work experimentally demonstrates this integration using IBM quantum computers, showcasing its feasibility with current technology. The results provide a foundation for scalable quantum networks, enabling advancements in quantum memory, secure authentication, and even quantum economics

QUANTUM ENERGY TELEPORTATION (QET)

QET is a theoretical framework where energy is transferred between quantum systems (e.g., qubits) without direct physical transportation or energy flow.
This QET leverages quantum entanglement to transfer energy between two distant locations without any physical movement of the energy



$$|g\rangle = \frac{1}{\sqrt{2}}\sqrt{1-\frac{h}{\sqrt{h^2+k^2}}}|00\rangle - \frac{1}{\sqrt{2}}\sqrt{1+\frac{h}{\sqrt{h^2+k^2}}}|11\rangle.$$

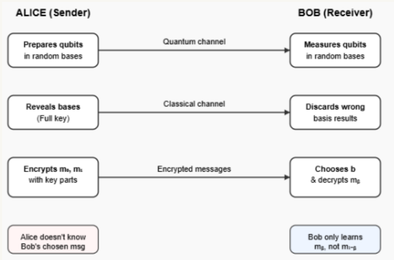
$$P_A(\mu) = \frac{1}{2}(1+\mu X_0),\ \mu \in \{-1,1\}$$

$$E_A = \sum_{\mu\in\{0,1\}} \frac{1}{p_A(\mu)} \langle g|P_A(\mu)H_A P_A(\mu)|g\rangle$$

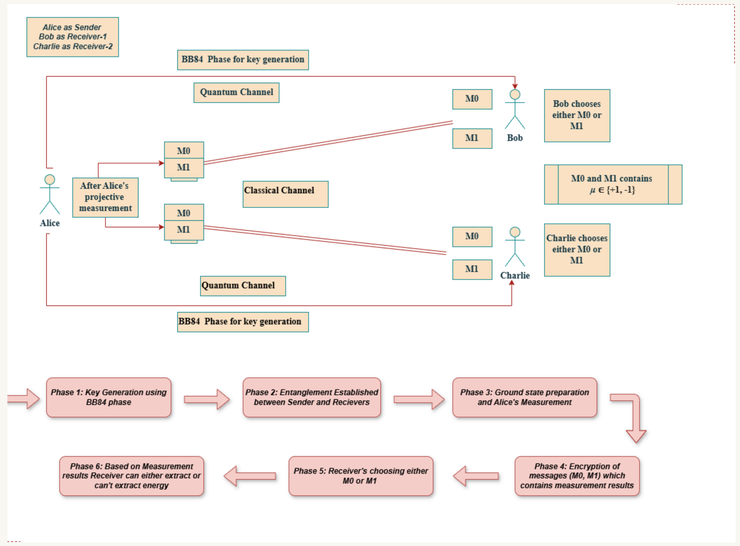
$$\langle E_B\rangle = \sum_{\mu\in\{-1,1\}} \langle g|P_A(\mu)U_B^{\dagger}(\mu)(H_B+V)U_B(\mu)P_A(\mu)|g\rangle$$
$$= -\frac{1}{\sqrt{h^2+k^2}}[hk\sin(2\phi) - (h^2+2k^2)(1-\cos(2\phi))],$$

QUANTUM OBLIVIOUS TRANSFER(QOT)

Quantum Oblivious Transfer (QOT) is a cryptographic protocol in which a Sender (Alice) has (at least) two different pieces of information, and a Receiver (Bob) can choose to learn exactly one of those pieces while remaining oblivious of the other.



METHODOLOGY

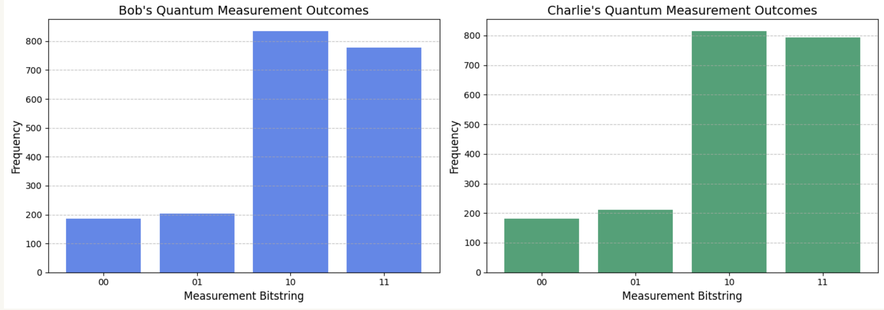


The methodology for our protocol is as below-
1. Following the QET protocol, Alice creates entanglement state $|\Phi+\rangle$ with her qubits along with Bob's and Charlie's independently after preparing $|g\rangle$ state.
2. Alice encodes her projective measurement results $\mu B, \mu C \in \{+1, -1\}$ into two messages (one correct & one incorrect).
3. Receivers choose one of the two messages following the QOT (BBCS) protocol. So, one of them or both or neither of them get the correct results.
4. Receivers use unitary operations on their qubits based on those correct results to extract energy, where incorrect results violate the QET protocol with no energy extraction.

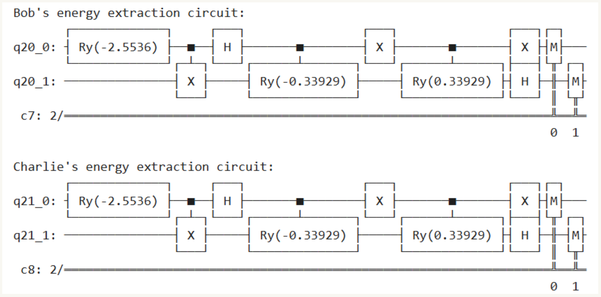
NOVELTY

Our protocol enhance the security of energy teleportation maintaining obliviousness between sender and receivers by integrating QOT with QET protocol. Correct results enable energy extraction while incorrect results fail which guarantees no biasness as sender doesn't get to control energy extraction of the receivers, which also creates obliviousness between receivers.

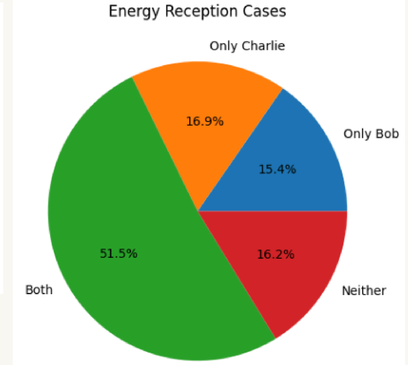
RESULTS AND ANALYSIS



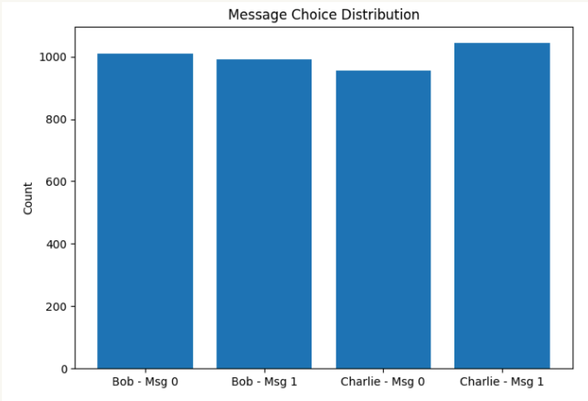
Our experimental results shows that the Energy Teleportation using Quantum Oblivious method was successful as in the graph the histograms show the distribution of two-qubit measurement outcomes (bitstrings) for Bob and Charlie



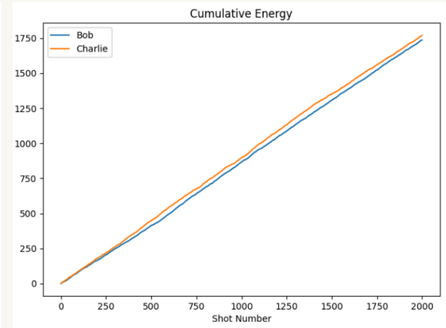
This is the circuit Diagram for Quantum Energy Teleportation after doing conditional operation based on Alice's measurements



Here in the pi-chart it is shown that Bob and Charlie can have 3 possibilities as They both can either get energy or one of them or neither of them can get the enery



Here in the graph four possible outcomes of two messages M0 &M1 is shown.



In the graph, it is shown that they almost get equal amount of energy, so neither of them get biased.

CONCLUSION

This protocol tries to establish a foundation for quantum secured oblivious energy teleportation by integrating QOT with QET. Though we've achieved significant results in a noise-free environment, future works may include experimental validation with different noise model and optimized error correction to make a connection with practical life energy distribution.

ACKNOWLEDGEMENT

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